

STAT 4005 - Time Series

Chapter 6 - Forecasting

Agenda

- 1 Introduction
- 2 Box and Jenkins approach
- 3 Summary: Time series analysis
- 4 Example: Treasury bill
- 5 Example: Accident death

Introduction

Forecast function

$$Y_{n+h}^n \stackrel{\text{def}}{=} E(Y_{n+h} | Y_n, Y_{n-1}, \dots, Y_1)$$

- Y_{n+h}^n is a function of the observation $(Y_1, \dots, Y_n)$ used for prediction
- Y_{n+h} is the unknown future value

Forecast error

- $e_n(h) = Y_{n+h} - Y_{n+h}^n$

Variance of forecast error

$$P_{n+h}^n = \text{Var}(e_n(h) | Y_n, \dots, Y_1) = E[(Y_{n+h} - Y_{n+h}^n)^2 | Y_n, \dots, Y_1]$$

95% prediction interval for Y_{n+h}

$$\left(Y_{n+h}^n - 1.96\sqrt{P_{n+h}^n}, Y_{n+h}^n + 1.96\sqrt{P_{n+h}^n} \right)$$

Box and Jenkins Approach (General (but tedious) Version)

Work for **causal** and **invertible** ARMA(p, q) model : $\phi(B)Y_t = \theta(B)Z_t$

- ❶ Causal representation: $Y_t = \sum_{i=0}^{\infty} \psi_i Z_{t-i}$
- ❷ Forecast: Using $E(Z_s | Y_n, \dots, Y_1) = Z_s \cdot 1_{\{s \leq n\}}$, we have

$$Y_{n+h}^n = E(Y_{n+h} | Y_n, \dots, Y_1) = E\left(\sum_{i=0}^{\infty} \psi_i Z_{n+h-i} \middle| Y_n, \dots, Y_1\right) = \sum_{i=h}^{\infty} \psi_i Z_{n+h-i}$$

- ❸ Forecast error: $e_n(h) = Y_{n+h} - Y_{n+h}^n = \sum_{i=0}^{h-1} \psi_i Z_{n+h-i}$
- ❹ Variance of forecast error

$$P_{n+h}^n = \text{Var}(e_n(h) | Y_n, \dots, Y_1) = \sigma^2 \sum_{i=0}^{h-1} \psi_i^2$$

- ❺ 95% prediction interval for Y_{n+h} :

$$Y_{n+h}^n \pm 1.96 \sqrt{P_{n+h}^n}$$

- Past noises Z_k , $k \leq n$ can be solved from the ARMA equation recursively by assuming that $Z_n = 0$ for $n \leq p$. (Invertibility needed!)
- Future noises Z_k , $k > n$ are unknown and are treated as 0 (their expectation)

Box and Jenkins Approach (Recursive Version)

Work for **causal** and **invertible** ARMA(p, q) model : $\phi(B)Y_t = \theta(B)Z_t$

- ❶ Recursively find $Z_k, k \leq n$ from $Y_1, \dots, Y_n$.
- ❷ Recursively find Y_{n+h}^n and $e_n(h) = Y_{n+h} - Y_{n+h}^n, h \geq 1$:

$$\begin{aligned}\bullet Y_{n+1} &= \phi_1 Y_n + \dots + \phi_p Y_{n+1-p} + Z_{n+1} - \theta_1 Z_n - \dots - \theta_q Z_{n+1-q} \\ Y_{n+1}^n &= \phi_1 Y_n + \dots + \phi_p Y_{n+1-p} + 0 - \theta_1 Z_n - \dots - \theta_q Z_{n+1-q} \\ e_n(1) &= Y_{n+1} - Y_{n+1}^n = Z_{n+1} . \\ \bullet Y_{n+2} &= \phi_1 Y_{n+1} + \dots + \phi_p Y_{n+2-p} + Z_{n+2} - \theta_1 Z_{n+1} - \dots - \theta_q Z_{n+2-q} \\ Y_{n+2}^n &= \phi_1 Y_{n+1}^n + \dots + \phi_p Y_{n+2-p} + 0 - 0 - \theta_2 Z_n - \dots - \theta_q Z_{n+2-q} \\ e_n(2) &= \phi_1 (Y_{n+1} - Y_{n+1}^n) + Z_{n+2} - \theta_1 Z_{n+1} = \phi_1 e_n(1) + Z_{n+2} - \theta_1 Z_{n+1} \\ \bullet Y_{n+3}^n &= \phi_1 Y_{n+2}^n + \phi_2 Y_{n+1}^n + \phi_3 Y_n + \dots + \phi_p Y_{n+3-p} - \theta_3 Z_n - \dots - \theta_q Z_{n+3-q} \\ e_n(3) &= \phi_1 e_n(2) + \phi_2 e_n(1) + Z_{n+3} - \theta_1 Z_{n+2} - \theta_2 Z_{n+1} \\ \dots &= \dots \\ \bullet Y_{n+h}^n &= \phi_1 Y_{n+h-1}^n + \phi_2 Y_{n+h-2}^n + \dots + \phi_p Y_{n+h-p}^n \\ e_n(h) &= \phi_1 e_n(h-1) + \phi_2 e_n(h-2) + \dots + \phi_p e_n(h-p) \\ &\quad + Z_{n+h} - \theta_1 Z_{n+h-1} - \dots - \theta_q Z_{n+h-q} . \quad (h > \max(p, q))\end{aligned}$$

- ❸ 95% Prediction interval: $Y_{n+h}^n \pm 1.96 \sqrt{\text{Var}(e_n(h) | Y_n, \dots, Y_1)}$

Box and Jenkins Approach: Example 1 (General Version)

Example : AR(1) : $Y_t = \alpha Y_{t-1} + Z_t$, $Z_t \sim N(0, \sigma^2)$

i) Causal Representation: $Y_t = \sum_{i=0}^{\infty} \alpha^i Z_{t-i}$.

ii) Forecast:

$$\begin{aligned} Y_{n+h}^n &= E(Y_{n+h} | Y_n, Y_{n-1}, \dots, Y_1) = E\left(\sum_{i=0}^{\infty} \alpha^i Z_{n+h-i} \middle| Y_n, \dots, Y_1\right) \\ &= \sum_{i=h}^{\infty} \alpha^i Z_{n+h-i} = \alpha^h \sum_{i=0}^{\infty} \alpha^i Z_{n-i} = \alpha^h Y_n. \end{aligned}$$

iii) Forecast error: $e_n(h) = Y_{n+h} - Y_{n+h}^n = \sum_{i=0}^{h-1} \alpha^i Z_{n+h-i}$

iv) Variance of forecast error:

$$P_{n+h}^n = \text{Var}(e_n(h)) = \sigma^2 \sum_{i=0}^{h-1} \alpha^{2i} = \sigma^2 \frac{1 - \alpha^{2h}}{1 - \alpha^2}.$$

v) 95% Prediction interval: $\alpha^h Y_n \pm 1.96\sigma \sqrt{\frac{1 - \alpha^{2h}}{1 - \alpha^2}}$

Box and Jenkins Approach: Example 1 (Recursive Version)

Example : AR(1) : $Y_t = \alpha Y_{t-1} + Z_t$, $Z_t \sim N(0, \sigma^2)$

i) 1-Step forecast

$$Y_{n+1}^n = E(Y_{n+1}|Y_n, \dots, Y_1) = E(\alpha Y_n + Z_{n+1}|Y_n, \dots, Y_1) = \alpha Y_n$$

$$e_n(1) = Y_{n+1} - Y_{n+1}^n = Z_{n+1}$$

$$P_{n+1}^n = \text{Var}(e_n(1)^2|Y_n, \dots, Y_1) = E(Z_{n+1}^2) = \sigma^2$$

ii) 2-Step forecast

$$Y_{n+2}^n = E(\alpha Y_{n+1} + Z_{n+2}|Y_n, \dots, Y_1) = \alpha Y_{n+1}^n = \alpha^2 Y_n$$

$$e_n(2) = Y_{n+2} - Y_{n+2}^n = \alpha(Y_{n+1} - Y_{n+1}^n) + Z_{n+2} = \alpha Z_{n+1} + Z_{n+2}$$

$$P_{n+2}^n = \text{Var}(e_n(2)^2|Y_n, \dots, Y_1) = \sigma^2(1 + \alpha^2)$$

iii) 95% Prediction intervals

- 1-Step: $\alpha Y_n \pm 1.96\sigma$
- 2-Step: $\alpha^2 Y_n \pm 1.96\sigma\sqrt{1 + \alpha^2}$
- Agree with general version: $\alpha^h Y_n \pm 1.96\sigma\sqrt{\frac{1 - \alpha^{2h}}{1 - \alpha^2}}$

Box and Jenkins Approach: Example 2

SARIMA(1, 0, 0) $\times$ (0, 1, 1)₁₂ Model:

$$Y_n - Y_{n-12} = \phi(Y_{n-1} - Y_{n-13}) + Z_n - \theta Z_{n-12}$$

Forecast:

- One-step: $Y_{n+1}^n = Y_{n-11} + \phi(Y_n - Y_{n-12}) - \theta Z_{n-11}$
- Two-step: $Y_{n+2}^n = Y_{n-10} + \phi(Y_{n+1}^n - Y_{n-11}) - \theta Z_{n-10}$

Forecast Error:

- One-step: $Y_{n+1} - Y_{n+1}^n = Z_{n+1}$
- Two-step: $Y_{n+2} - Y_{n+2}^n = Z_{n+2} + \phi(Y_{n+1} - Y_{n+1}^n) = Z_{n+2} + \phi Z_{n+1}$

Variance of forecast error:

- One-step: $P_{n+1}^n = \text{Var}(Z_{n+1}) = \sigma^2$
- Two-step: $P_{n+2}^n = \text{Var}(Z_{n+2} + \phi Z_{n+1}) = (1 + \phi^2)\sigma^2$

95% Prediction Intervals:

- One-step: $Y_{n+1}^n \pm 1.96\sqrt{P_{n+1}^n} = Y_{n+1}^n \pm 1.96\sigma$
- Two-step: $Y_{n+2}^n \pm 1.96\sqrt{P_{n+2}^n} = Y_{n+2}^n \pm 1.96\sigma\sqrt{1 + \phi^2}$

Box and Jenkins Approach: Example 3

- ARMA(2,2) model (under the specifications in R)

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}$$

- Assume that $Z_n = 0$ for $n \leq p = 2$, we have,

$$\begin{aligned} Y_3 &= \phi_1 Y_2 + \phi_2 Y_1 + Z_3 && \Rightarrow \text{get } Z_3 \\ Y_4 &= \phi_1 Y_3 + \phi_2 Y_2 + Z_4 + \theta_1 Z_3 && \Rightarrow \text{get } Z_4 \\ Y_5 &= \phi_1 Y_4 + \phi_2 Y_3 + Z_5 + \theta_1 Z_4 + \theta_2 Z_3 && \Rightarrow \text{get } Z_5 \end{aligned}$$

- Recursively get $Z_6, Z_7, \dots, Z_n$

$$\begin{aligned} Y_{n+1}^n &= \phi_1 Y_n + \phi_2 Y_{n-1} + \theta_1 Z_n + \theta_2 Z_{n-1} \\ Y_{n+2}^n &= \phi_1 Y_{n+1}^n + \phi_2 Y_n + \theta_2 Z_n \\ Y_{n+h}^n &= \phi_1 Y_{n+h-1}^n + \phi_2 Y_{n+h-2}^n \end{aligned}$$

Box and Jenkins Approach: Example 3

R Codes:

```
set.seed(123)
n=200
x=arima.sim(n,model=list(ar=c(0.5,0.2),ma=c(0.1,0.6)))
fit=arima(x,order=c(2,0,2),include.mean=F)
p=fit$coef
Z=rep(0,n)
for (j in 3:n){
  Z[j]=x[j]-p[1]*x[j-1]-p[2]*x[j-2]-p[3]*Z[j-1]-p[4]*Z[j-2] }
Pred1=p[1]*x[n]+p[2]*x[n-1]+p[3]*Z[n]+p[4]*Z[n-1]
Pred2=p[1]*Pred1+p[2]*x[n]+p[4]*Z[n]
Pred1;Pred2
predict(fit,n.ahead=2)
```

Exercise

Let $Y_t = Z_t + 0.4Z_{t-1}$, $Z_t \stackrel{\text{iid}}{\sim} N(0, 4)$. Given $Y_1 = 0.5$, $Y_2 = 2.1$, $Y_3 = 1.2$, $Y_4 = -0.8$.

- (a) Using all the information up to day 4, predict Y_5 . Find the prediction error and 95% prediction interval.
- (b) Using all the information up to day 4, predict Y_6 . Find the prediction error and 95% prediction interval.
- (c) Using all the information up to day 4, predict Y_{100} . Find the prediction error and 95% prediction interval.

Solution I

a) First, we compute Z_1, Z_2, Z_3, Z_4 given $Z_0 = 0$ using Y_1, Y_2, Y_3, Y_4 .

$$Z_1 = Y_1 = 0.5$$

$$Z_2 = Y_2 - 0.4Z_1 = 1.9$$

$$Z_3 = Y_3 - 0.4Z_2 = 0.44$$

$$Z_4 = Y_4 - 0.4Z_3 = -0.976.$$

The one-step ahead forecast is

$$\begin{aligned} Y_5^4 &= E(Y_5|Y_1, Y_2, Y_3, Y_4) \\ &= E(Z_5 + 0.4Z_4|Y_1, Y_2, Y_3, Y_4) \\ &= 0.4E(Z_4|Y_1, Y_2, Y_3, Y_4) \\ &= 0.4Z_4 = -0.3904. \end{aligned}$$

Solution II

The one-step ahead forecast error is

$$e_4(1) = Y_5 - Y_5^4 = Z_5.$$

The forecast error variance is given by

$$E(e_4(1)^2 | Y_1, Y_2, Y_3, Y_4) = E(Z_5^2) = 4.$$

Since Z_t is assumed to be normally distributed, we have $e_4(1) \sim N(0, 4)$. Thus, the prediction interval for the value Y_5 is constructed as

$$(Y_5^4 - \sqrt{4} \times 1.96, Y_5^4 + \sqrt{4} \times 1.96) = (-4.31, 3.53).$$

Solution III

- First, we have the two-step ahead forecast

$$Y_6^4 = E(Z_6 + 0.4Z_5|Y_1, Y_2, Y_3, Y_4) = 0,$$

and forecast error

$$e_4(2) = Y_6 - Y_6^4 = Z_6 + 0.4Z_5.$$

Therefore, the forecast error variance is

$$\text{Var}(e_4^2(2)|Y_1, Y_2, Y_3, Y_4) = E(e_4^2(2)|Y_1, Y_2, Y_3, Y_4) = 1.16 \times 4 = 4.64,$$

so the prediction interval is constructed as

$$(Y_6^4 - \sqrt{4.64} \times 1.96, Y_6^4 + \sqrt{4.64} \times 1.96) = (-4.22, 4.22).$$

- e) The forecast is $Y_{100}^4 = 0$ and the forecast error is $e_4(96) = Y_{100} = Z_{100} + 0.4Z_{99}$. The variance of the forecast error is

$$E(e_4(96))^2 | Y_1, Y_2, Y_3, Y_4 = 1.16 \times 4.$$

So the prediction interval for Y_{100} is the same as that of Y_6 , i.e. $(-4.22, 4.22)$.

Exercise

Let $Y_t = 0.1Y_{t-1} + 0.2Y_{t-2} + Z_t$ with $Z_t \stackrel{\text{iid}}{\sim} N(0, 25)$. Let $Y_{199} = 120$, $Y_{200} = 100$.

- (a) Find the forecasts Y_{201}^{200} and Y_{202}^{200} and Y_{205}^{200}
- (b) Find the 95% prediction interval of Y_{201} and the 95% prediction interval of Y_{203} .
- (c) Update your forecast for Y_{205} given $Y_{201} = 11$.
- (d) Find $\text{Cov}[e_t(m), e_t(k)]$, where m and k are both positive integers.

Solution I

a) Using the Box and Jenkins forecasting approach, we have

$$Y_{201}^{200} = E(Y_{201}|Y_{200}) = 0.1Y_{200} + 0.2Y_{199} = 34$$

$$Y_{202}^{200} = E(Y_{202}|Y_{200}) = 0.1Y_{201}^{200} + 0.2Y_{200} = 23.4$$

$$Y_{203}^{200} = E(Y_{203}|Y_{200}) = 0.1Y_{202}^{200} + 0.2Y_{201}^{200} = 9.14$$

$$Y_{204}^{200} = E(Y_{204}|Y_{200}) = 0.1Y_{203}^{200} + 0.2Y_{202}^{200} = 5.6$$

$$Y_{205}^{200} = E(Y_{205}|Y_{200}) = 0.1Y_{204}^{200} + 0.2Y_{203}^{200} = 2.39.$$

Solution II

- The variance of forecast error is

$$\begin{aligned} & \text{Var}(e_{200}(1)|Y_{200}, \dots, Y_1) \\ &= E(e_{200}(1)^2|Y_{200}, \dots, Y_1) \\ &= E[(Y_{201} - Y_{201}^{200})^2|Y_{200}, \dots, Y_1] \\ &= E(Z_{201}^2|Y_{200}, \dots, Y_1) \\ &= E(Z_{201}^2) \\ &= 25. \end{aligned}$$

Thus, the prediction interval of Y_{201} can be constructed as

$$(Y_{201}^{200} - 1.96\sqrt{25}, Y_{201}^{200} + 1.96\sqrt{25}) = (24.2, 43.8).$$

Solution III

For the 95% prediction interval of Y_{203} , we need to find the forecast error of Y_{202}^{200} , then find the forecast error of Y_{203}^{200} . First, as $e_{200}(1) = Y_{201} - Y_{201}^{200} = Z_{201}$, the forecast error of Y_{202}^{200} is

$$\begin{aligned}e_{200}(2) &= Y_{202} - Y_{202}^{200} \\&= Z_{202} + 0.1(Y_{201} - Y_{201}^{200}) \\&= Z_{202} + 0.1Z_{201} .\end{aligned}$$

Next, forecast error of Y_{203}^{200} is

$$\begin{aligned}e_{200}(3) &= Y_{203} - Y_{203}^{200} \\&= Z_{203} + 0.1(Y_{202} - Y_{202}^{200}) + 0.2(Y_{201} - Y_{201}^{200}) \\&= Z_{203} + 0.1e_{200}(2) + 0.2e_{200}(1) \\&= Z_{203} + 0.1Z_{202} + 0.21Z_{201} .\end{aligned}$$

Solution IV

Hence, the variance of forecast is

$$\begin{aligned} & \text{Var}(e_{200}(3)^2 | Y_{200}, \dots, Y_1) \\ &= E[(Z_{203} + 0.1Z_{202} + 0.21Z_{201})^2 | Y_{200}, \dots, Y_1] \\ &= E(Z_{203}^2) + 0.1^2 E(Z_{202}^2) + 0.21^2 E(Z_{201}^2) \\ &= (1 + 0.01 + 0.0441) \times 25 \\ &= 26.35 \end{aligned}$$

Finally, the 95% prediction interval for Y_{203} is constructed as

$$(Y_{203}^{200} - 1.96\sqrt{26.35}, Y_{203}^{200} + 1.96\sqrt{26.35}) = (-0.92, 19.2).$$

Solution V

- c) Following the steps in part (a), we have

$$Y_{202}^{201} = 0.1Y_{201} + 0.2Y_{200} = 21.1$$

$$Y_{203}^{201} = 0.1Y_{202}^{201} + 0.2Y_{201} = 4.31$$

$$Y_{204}^{201} = 0.1Y_{203}^{201} + 0.2Y_{202}^{201} = 4.65$$

$$Y_{205}^{201} = 0.1Y_{204}^{201} + 0.2Y_{203}^{201} = 1.327.$$

- d) Rewrite the AR model as

$$Z_t = (1 + 0.4B)(1 - 0.5B)Y_t.$$

Solution VI

The causal (MA) representation is given by

$$\begin{aligned} Y_t &= (1 + (-0.4)B + (-0.4)^2 B^2 + \dots)(1 + 0.5B + 0.5^2 B^2 + \dots)Z_t \\ &= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} (-0.4)^j 0.5^k B^{j+k} Z_t \\ &= \sum_{i=0}^{\infty} \left(\sum_{j \geq 0, k \geq 0, j+k=i} (-0.4)^j 0.5^k \right) B^i Z_t \\ &= \sum_{i=0}^{\infty} \left(\sum_{j=0}^i (-0.4)^j 0.5^{i-j} \right) B^i Z_t \\ &= \sum_{i=0}^{\infty} \psi_i Z_{t-i}, \end{aligned}$$

where $\psi_i = \sum_{j=0}^i (-0.4)^j (0.5)^{i-j} = (0.5)^i \frac{1 - (-0.8)^{i+1}}{1.8}$. Therefore, the forecast error is

$$e_t(k) = Y_{t+k} - Y_{t+k}^t = \sum_{i=0}^{\infty} \psi_i Z_{t+k-i} - \sum_{i=k}^{\infty} \psi_i Z_{t+k-i} = \sum_{i=0}^{k-1} \psi_i Z_{t+k-i}.$$

Solution VII

Without loss of generality, suppose $k \geq m$, we have

$$\begin{aligned} & \text{Cov}(e_t(k), e_t(m)) \\ = & \sigma^2 \sum_{i=0}^{m-1} \psi_i \psi_{i+k-m} \\ = & 25 \sum_{i=0}^{m-1} (0.5)^i \frac{1 - (-0.8)^{i+1}}{1.8} (0.5)^{i+k-m} \frac{1 - (-0.8)^{i+k-m+1}}{1.8} \\ = & 25 \frac{0.5^{k-m}}{1.8^2} \sum_{i=0}^{m-1} (0.5)^{2i} (1 - (-0.8)^{i+1})(1 - (-0.8)^{i+k-m+1}) \\ = & 25 \frac{0.5^{k-m}}{1.8^2} \sum_{i=0}^{m-1} \left(0.25^i + 0.8(-0.2)^i + 0.8(-0.8)^{k-m}(-0.2)^i \right. \\ & \quad \left. + (-0.8)^{k-m+2}(0.16)^i \right) \\ = & 25 \frac{0.5^{k-m}}{1.8^2} \left(\frac{1 - 0.25^m}{0.75} + 0.8((-0.8)^{k-m} + 1) \frac{1 - (-0.2)^m}{1.2} \right. \\ & \quad \left. + (-0.8)^{k-m+2} \frac{1 - 0.16^m}{0.84} \right). \end{aligned}$$

Time Series Analysis: Standard Procedure

① Exploratory Data Analysis

- ① TS/ACF/PACF **plots** for observations Y_t
- ② Estimate/Remove trend and seasonal effect by **least squares/filtering/differencing** $\Rightarrow$ obtain noise Z_t for further analysis

② Model Identification/Fitting

- ① TS/ACF/PACF **plots** for noise Z_t to ensure stationarity
- ② Fit models by **MM, YW, LSE, CLS, MLE...** and select model by **FPE/AIC/BIC...**

③ Model Checking/Diagnostics

- ① Residual analysis (TS/ACF/PACF **plots** for residuals $\hat{e}_t = Z_t - \hat{Z}_t$)
- ② **Ljung-Box test**

④ Prediction

- ① **Box-Jenkins Approach**: Z_{n+h}^n ; Prediction Interval for Z_{n+h}
- ② Add trend & Seasonal effect/Integrate to obtain Y_{n+h}^n and P.I.

⑤ Output:

- ① Time Series = Trend + Seasonal effect + ARMA model
- ② Time Series = ARIMA model

Treasury Bill Example

- Monthly interest rate on three-month government Treasury bills from 1950-1988
- `ustbill.dat`
- Exploratory Data Analysis:

R-code:

```
ttbill=read.table("D://ustbill.dat")  
ttbill=ttbill[,-1]  
ttbill=as.vector(t(ttbill))
```

```
par(mfrow=c(2,2))  
plot.ts(ttbill)  
acf(ttbill)  
plot.ts(log(ttbill))  
acf(log(ttbill))
```

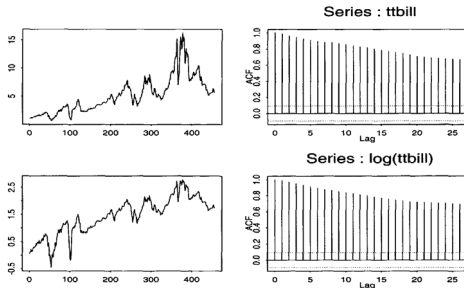


Fig. 6.1 Time series plots, ACF of T-bills, and log(T-bills).

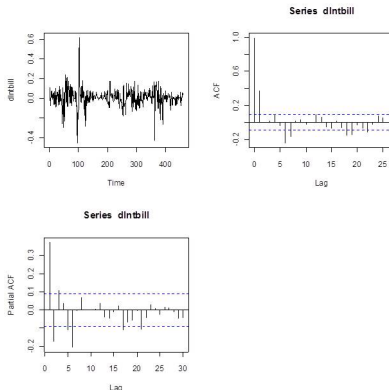
Treasury Bill Example

- From the plot, the data are nonstationary in both mean and variance
- Consider the difference in the log of the data

R-code:

```
lntbill=log(ttbill)  
dlntbill=diff(lntbill)
```

```
par(mfrow=c(2,2))  
plot.ts(dlntbill)  
acf(dlntbill)  
pacf(dlntbill,30)
```



- Both ACF and PACF suggest strong correlations at lags 1,6
- Try AR(6), MA(6) and ARMA(6,6) (AIC and diagnostic tests)

Fitting of AR(6)

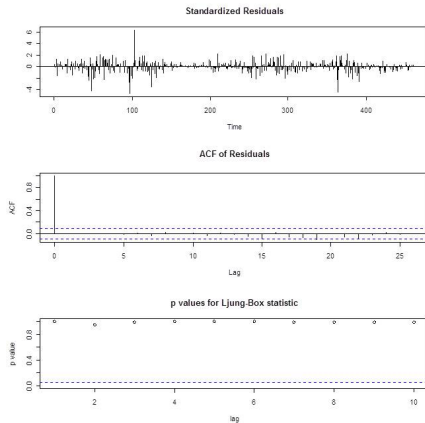
R-code:

```
d3=arima(dlnrtbill,order=c(6,0,0))
```

```
tsdiag(d3)
```

```
d3$aic
```

AIC = -1056.251



Fitting of MA(6)

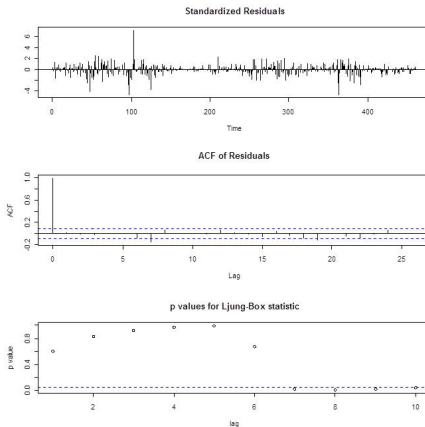
R-code:

```
d4=arima(dlntbill,order=c(0,0,6))
```

```
tsdiag(d4)
```

```
d4$aic
```

AIC = -1038.066



Fitting of ARMA(6,6)

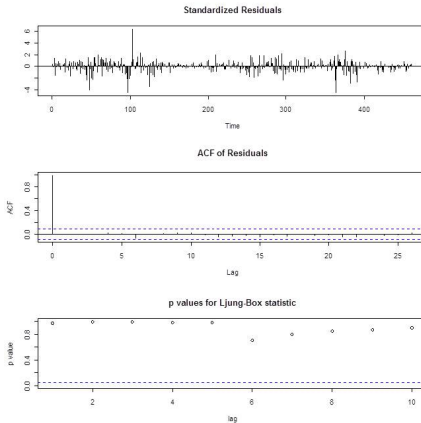
R-code:

```
d5=arima(dlntbill,order=c(6,0,6))
```

```
tsdiag(d5)
```

```
d5$aic
```

AIC = -1055.556



Treasury Bill Example

From AIC and diagnostic tests, AR(6) seems to be the best

$$\text{Model: } X_t = \log(\text{tbill}), Y_t = (1 - B)X_t$$

$$(1 - 0.44B + 0.19B^2 - 0.09B^3 - 0.04B^4 + 0.01B^5 + 0.21B^6)Y_t = 0.0039 + Z_t$$

Forecasting procedure:

1. Fit an AR(6) to $\{Y_t\}_{t=1,\dots,n}$ and obtain forecasts $\hat{Y}_{n+h}$
2. Recover $\hat{X}_{t+h}$ from $\hat{Y}_t$ by "integrating":

$$\hat{X}_{n+h} = \hat{Y}_{n+h} + \hat{X}_{n+h-1}$$

3. Recover $\widehat{\text{tbill}}_{n+h}$ as

$$\begin{aligned}\widehat{\text{tbill}}_{n+h} &= \exp(\hat{X}_{n+h}) = \exp(\hat{Y}_{n+h}) \exp(\hat{X}_{n+h-1}) \\ &= \exp(\hat{Y}_{n+h}) \widehat{\text{tbill}}_{n+h-1}\end{aligned}$$

Treasury Bill Example

Model: $X_t = \log(\text{tbill}), Y_t = (1 - B)X_t, Y_t \sim \text{AR}(6)$

Forecast: $\widehat{\text{tbill}}_{n+h} = \exp(\widehat{Y}_{n+h})\widehat{\text{tbill}}_{n+h-1}$

R-code: (Use days 1 to 456 to forecast days 457 to 462)

- fit=arima(dlnrtbill[1:455],order=c(6,0,0)) # 455th differenced is 456th real time
Y=predict(fit,n.ahead=6)

```
tfore=rep(0,6)
tfore[1]=exp(Y$pred[1])*ttbill[456]
for( i in 2:6){
  tfore[i]=exp(Y$pred[i])*tfore[i-1]
}
```

tfore: 5.889805 5.820464 5.708289 5.759812 5.910892 5.9551662

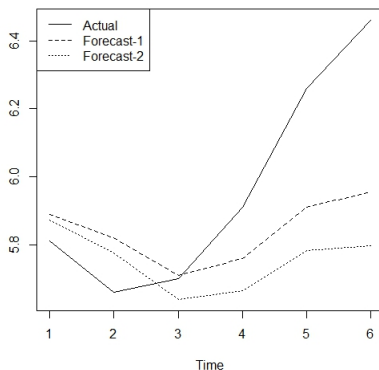
- fit2=arima(lntbill[1:456],order=c(6,1,0))

```
tfore2=exp(predict(fit2,n.ahead=6)$pred[1:6])
```

tfore2: 5.871463 5.776399 5.639398 5.663790 5.783611 5.797992 (Try include.mean=F for dlnrtbill)

- ttbill=as.ts(ttbill);tfore=as.ts(tfore);tfore2=as.ts(tfore2)
ts.plot(ttbill[457:462],tfore,tfore2,lty=c(1:3)) # 1-solid, 2-dashed, 3-dotted
leg.names=c("Actual","Forecast-1","Forecast-2") # Draw legend
legend("topleft",leg.names,lty=c(1:3))

Treasury Bill Example



- The forecasts for the first four months are reasonably well
- The later forecasts get worse as the prediction variability grows with time

Treasury Bill Example: Prediction Intervals

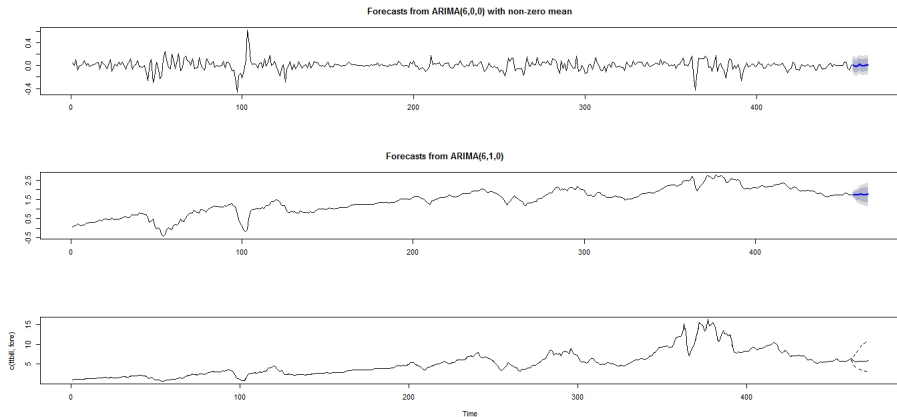
- Model: $X_t = \log(\text{tbill})$, $Y_t = (1 - B)X_t$, $Y_t \sim \text{AR}(6)$
- Forecast: $\widehat{\text{tbill}}_{n+h} = \exp(\widehat{Y}_{n+h})\widehat{\text{tbill}}_{n+h-1} = \exp(\widehat{Y}_{n+h} + \cdots + \widehat{Y}_{n+1})\text{tbill}_n$
- P.I. for Y_{n+h} : $(\widehat{Y}_{n+h} \pm 1.96\sqrt{\text{Var}(e_{Y,n}(h))})$
- P.I. for X_{n+h} :
 - $\widehat{X}_{n+h} = \widehat{Y}_{n+h} + \widehat{X}_{n+h-1} = \cdots = \widehat{Y}_{n+h} + \cdots + \widehat{Y}_{n+1} + X_n$
 - $e_{X,n}(h) = X_{n+h} - \widehat{X}_{n+h} = \sum_{j=1}^h (Y_{n+j} - \widehat{Y}_{n+j}) = \sum_{j=1}^h e_{Y,n}(j)$
 - P.I.: $(\widehat{X}_{n+h} \pm 1.96\sqrt{\text{Var}(e_{X,n}(h))})$ (Tedious but doable)
- P.I. for tbill :

$$\left(e^{\widehat{X}_{n+h} - 1.96\sqrt{\text{Var}(e_{X,n}(h))}}, e^{\widehat{X}_{n+h} + 1.96\sqrt{\text{Var}(e_{X,n}(h))}} \right)$$

```
library(forecast)
par(mfrow=c(3,1))
plot(forecast(fit));plot(forecast(fit2))
```

```
fore=exp(as.vector(forecast(fit2)$mean))
lower=exp(forecast(fit2)$lower[,2]); upper=exp(forecast(fit2)$upper[,2])
ts.plot(c(ttbill,fore)); lines(c(ttbill,lower),lty=2); lines(c(ttbill,upper),lty=2)
```

Treasury Bill Example: Prediction Intervals



- Top: Forecasts for $Y_t \sim AR(6)$
- Middle: Forecasts for X_t where $Y_t = (1 - B)X_t$
- Bottom: Forecasts for $tbill_t = e^{X_t}$

Accident Death Example

Number of accident deaths on U.S. highways between 1973 and 1978

- `accdeaths.dat`

- Read data:

R-code:

```
par(mfrow=c(2,2))
```

```
accdeaths=scan("D://accdeaths.dat")
```

```
ts.plot(accdeaths)
```

```
acf(accdeaths,30)
```

```
pacf(accdeaths,30)
```

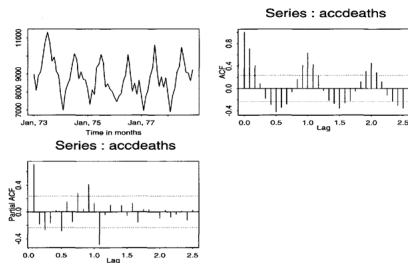


Fig. 5.3 Time series plots, ACF, and PACF of `accdeaths`.

- The ACF graph shows a periodic component (seasonal pattern) at lag 12, 24, 36, ...

Accident Death Example: Estimating Trend and Seasonal Effect

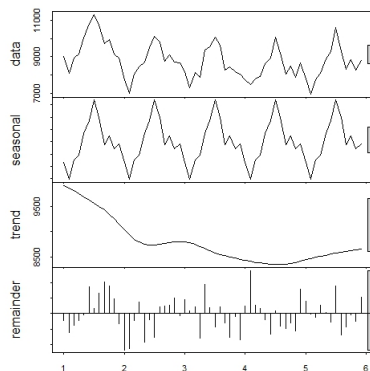
- Data: 1973-1978 ($n = 72$)
- Use 1973-1977 for model fitting, 1978 for evaluation
- R: automated decomposition of trend and seasonal effect

```
stl(data,s.window="periodic")
```

R-code:

```
test=accdeaths[61:72]      # 1978  
x=ts(accdeaths[1:60],frequency=12)  
# 1973-77  
decomp=stl(x,s.window="periodic")  
plot(decomp)
```

```
# Store the results  
trend=decomp$time.series[,2]  
seasonal=decomp$time.series[,1]  
y=decomp$time.series[,3]    # Noise
```



Accident Death Example: ARMA model fitting

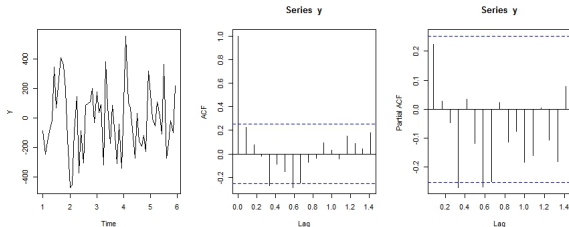
- After filtering out trend and seasonal effect, fit ARMA model for noise
- R: automated ARMA model selection and parameter estimation

```
auto.arima(data,ic=c("aicc"))           # "aic" or "bic"
```

```
library(forecast)
par(mfrow=c(1,3));ts.plot(y);acf(y);pacf(y)
arma.fit=auto.arima(y,ic=c("aic"),max.p=3,max.q=3,stepwise=F)
```

	ar1	ar2	ar3	ma1
	1.0323	-0.1086	-0.2014	-0.9603
s.e.	0.1307	0.1845	0.1289	0.0586

sigma^2 estimated as 39182: log likelihood=-403.19 AIC=816.39 AICc=817.5 BIC=826.86



- ARMA(3,1): $Y_t = 1.03Y_{t-1} - 0.11Y_{t-2} - 0.20Y_{t-3} + Z_t - 0.96Z_{t-1}$,
 $Z_t \sim N(0, 39182)$.

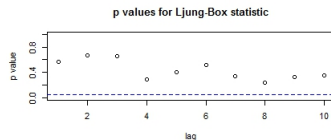
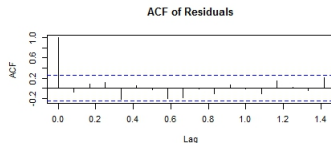
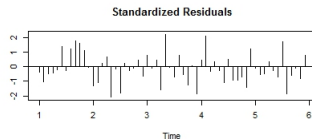
Accident Death Example: Model diagnostic

- R: Model diagnostic test and plots

```
tsdiag(arma.fit)
```

```
library(tseries)  
tsdiag(arma.fit)
```

- ACF-pass!
- Ljung-Box test-pass!



Accident Death Example: Prediction

- Prediction of Trend: (an naive way)

① $\hat{T}_{n+h} = T_{n+h}^n = T_n + h(T_n - T_{n-1})$

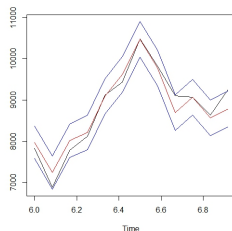
- Prediction of Seasonal Effect: use the suitable $S_1, \dots, S_{12}$.

- Prediction of ARMA: $\hat{Y}_{n+h} = Y_{n+h}^n \left(\pm 1.96 \sqrt{\text{Var}(e_n(h))} \right)$

Final prediction

$$\text{AccDeath}_{n+h} = \hat{T}_{n+h} + \hat{S}_{n+h} + \hat{Y}_{n+h} \quad \left(\pm 1.96 \sqrt{\text{Var}(e_n(h))} \right)$$

```
pred=predict(arma.fit,n.ahead=12)
inc=trend[60]-trend[59]
fore=trend[60]+inc*(1:12)+seasonal[1:12]+pred$pred
fore.up=fore+1.96*pred$se
fore.low=fore-1.96*pred$se
ts.plot(cbind(test,fore,fore.up,fore.low),col=c(1,2,4,4))
```

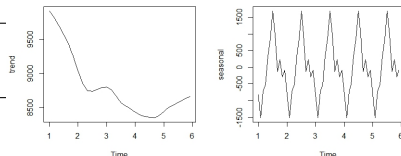


Accident Death Example: Summary

$$\text{AccDeath}_t = T_t + S_t + Y_t$$

- Trend and Seasonal Effects:

```
par(mfrow=c(1,2))  
ts.plot(trend); ts.plot(seasonal)
```



- ARMA(3,1) model

$$Y_t = 1.03Y_{t-1} - 0.11Y_{t-2} - 0.20Y_{t-3} \\ + Z_t - 0.96Z_{t-1}, \quad Z_t \sim N(0, 39182)$$

- Prediction

```
data=c(x,test)  
fore=c(x,fore)  
up=c(x,fore.up); low=c(x,fore.low)  
ts.plot(cbind(fore,up,low,data),col=c(2,4,4,1))
```

